Kronecker Products, vec, and Matrix Derivatives

Linear Algebra · Lecture 14

Last time (Lecture 13): the singular value decomposition $A = P\,\mathrm{diag}(D, 0)Q'$, the best approximation of rank k (Eckart–Young), principal components through the SVD, the spectral norm and the condition number. Throughout, I and $e_j = \mathrm{col}_j(I)$ are of the size required.

Facts

From calculus, used without proof (balls, open sets and partial derivatives: Definition 7).

F1. A function on an open set is C^1 (**continuously differentiable**) if its partial derivatives exist and are continuous there, C^2 if moreover the partial derivatives of these exist and are continuous; vector and matrix functions are C^1 (C^2) if all their entries are. Sums, products and quotients (nonzero denominator) of C^1 (C^2) functions are C^1 (C^2), with $\partial(f+g)/\partial x_j = \partial f/\partial x_j + \partial g/\partial x_j$, $\partial(fg)/\partial x_j = (\partial f/\partial x_j)g + f\,\partial g/\partial x_j$, $\partial(cf)/\partial x_j = c\,\partial f/\partial x_j$ (c constant); constants and the variables are C^2, so polynomials in the variables are C^2. A C^1 function is continuous, and a continuous function that is $\neq 0$ (> 0) at c is $\neq 0$ (> 0) on some ball around c. **Chain rule**: if g is C^1 on an open $S \subseteq \mathbb{R}^n$ with values in an open $T \subseteq \mathbb{R}^m$, and f is C^1 (C^2) on T, then $f \circ g$ is C^1 on S (C^2 if g is too) and $\mathrm{D}(f \circ g)(x) = \mathrm{D}f(g(x))\,\mathrm{D}g(x)$. $\log t$ ($t > 0$) is C^2, with derivative $1/t$.

F2. If f is C^2 on an open set, then $\partial^2 f/\partial x_i \partial x_j = \partial^2 f/\partial x_j \partial x_i$, where $\partial^2 f/\partial x_i \partial x_j$ is the partial derivative with respect to x_i of $\partial f/\partial x_j$.

F3. (One variable.) **Fermat's rule**: if h is differentiable on an open interval and has a local minimum or maximum at a point t_0 of it, then $h'(t_0) = 0$. **Taylor's formula**: if h is C^2 on an open interval that contains $[0, 1]$, then $h(1) = h(0) + h'(0) + \frac{1}{2}h''(\tau)$ for some $0 < \tau < 1$.

From earlier lectures (restated).

- Lecture 1, Theorem 3(a), Definitions 7, 9, Theorems 8, 10(a),(b): matrix products are associative and distributive, $c(AB) = (cA)B = A(cB)$; $(A')' = A$, $(A+B)' = A' + B'$, $(cA)' = cA'$, $(AB)' = B'A'$, $x'y = y'x$, and a 1×1 matrix (a number) is its own transpose; A is **symmetric** if $A' = A$; $\mathrm{tr}\,A = \sum_i a_{ii}$, $\mathrm{tr}(A+B) = \mathrm{tr}\,A + \mathrm{tr}\,B$, $\mathrm{tr}(cA) = c\,\mathrm{tr}\,A$, $\mathrm{tr}(AB) = \mathrm{tr}(BA)$ ($A \in \mathbb{R}^{m\times n}$, $B \in \mathbb{R}^{n\times m}$). Lecture 1, Definitions 12, 14: $\mathrm{diag}(d_1, \ldots, d_n)$ has the diagonal entries $d_1, \ldots, d_n$ and all other entries 0; the **summing vector** $\mathbf{1}_n \in \mathbb{R}^n$ has all entries 1.
- Lecture 1, Theorem 4(a),(b),(d): for $A = [a_1\ \cdots\ a_n] \in \mathbb{R}^{m\times n}$, $B = [b_1\ \cdots\ b_p] \in \mathbb{R}^{n\times p}$ and $x \in \mathbb{R}^n$: $\mathrm{col}_j(AB) = Ab_j$; $Ax = x_1a_1 + \cdots + x_na_n$, $Ae_j = a_j$; $(AB)_{ij} = \mathrm{row}_i(A)\,b_j$ and $AB = a_1\,\mathrm{row}_1(B) + \cdots + a_n\,\mathrm{row}_n(B)$ (the **outer product expansion**). Lecture 1, Proposition 11(a),(b): $e_i'Ae_j = a_{ij}$; for $A, B \in \mathbb{R}^{m\times n}$, $A = B \iff Ax = Bx$ for every $x \in \mathbb{R}^n$.
- Lecture 2, Definition 10, Theorem 15, Corollary 12: $A \in \mathbb{R}^{n\times n}$ is **invertible** if $AB = BA = I_n$ for some B, the **inverse** A^{-1}. If $BA = I$ or $AB = I$, then A is invertible and $B = A^{-1}$. If A is symmetric and invertible, A^{-1} is symmetric.
- Lecture 5, Definition 1, Theorem 2(b),(c): for a matrix A partitioned into blocks A_{ij}, A' has the (j,i) block A_{ij}'; for a partition of B conformal for multiplication, $(AB)_{ik} = A_{i1}B_{1k} + \cdots + A_{is}B_{sk}$.
- Lecture 5, Definition 7, Theorems 12(a), 13(a), 14: $\det(a) = a$ for $n = 1$; for $A \in \mathbb{R}^{n\times n}$, $n \geqslant 2$, $A(i|j)$ is A without row i and column j, $\det A(i|j)$ a **minor**, $c_{ij} = (-1)^{i+j}\det A(i|j)$ a **cofactor**, and $\det A = a_{11}c_{11} + \cdots + a_{n1}c_{n1}$. A is invertible $\iff \det A \neq 0 \iff \mathrm{rank}\,A = n$; $\det A = a_{i1}c_{i1} + \cdots + a_{in}c_{in}$ for every i (expansion along row i); the **adjugate** $\mathrm{adj}\,A = (c_{ij})'$ has $A\,\mathrm{adj}\,A = (\det A)I_n$, and $A^{-1} = \frac{1}{\det A}\,\mathrm{adj}\,A$ if $\det A \neq 0$.
- Lecture 6, Definition 1: the **length** $\|x\| = \sqrt{x'x}$, the **distance** $\|x - y\|$; $\|cx\| = |c|\,\|x\|$ ($c \in \mathbb{R}$). Lecture 13, Definition 2: the **Frobenius norm** $\|A\|_F = \left(\mathrm{tr}(A'A)\right)^{1/2} = \left(\sum_i \sum_j a_{ij}^2\right)^{1/2}$.
- Lecture 8, Definition 13, Theorems 14(c), 15(a),(b): a symmetric A is **positive definite** (PD) if $x'Ax > 0$ for all $x \neq 0$, **positive semidefinite** (PSD) if $x'Ax \geqslant 0$ for all x; a PD matrix is

PSD and invertible. If $B \in \mathbb{R}^{n\times n}$ is PD and $K \in \mathbb{R}^{n\times k}$ has $\operatorname{rank} K = k$, then $K'BK$ is PD. For $A \in \mathbb{R}^{n\times p}$, $A'A$ is PSD, and PD $\iff \operatorname{rank} A = p$.

- Lecture 11, Definition 1, Theorems 2(b), 3(a): in the linear model $y = X\beta + \varepsilon$ ($X \in \mathbb{R}^{n\times p}$, $y \in \mathbb{R}^n$), a **least-squares estimate** $\hat\beta$ minimizes $S(\beta) = \|y - X\beta\|^2$; $X'X\beta = X'y$ are the **normal equations**. $X'X$ is invertible $\iff \operatorname{rank} X = p$, and then $\hat\beta = (X'X)^{-1}X'y$ is their only solution. $\hat\beta$ is a least-squares estimate $\iff X'X\hat\beta = X'y$. Lecture 11, Example 5: $X = [\mathbf{1}_4\ (2,0,2,0)'\ (3,3,1,1)'] \in \mathbb{R}^{4\times 3}$, $y = (9,3,5,3)'$; $X'X = \left(\begin{smallmatrix} 4 & 4 & 8 \\ 4 & 8 & 8 \\ 8 & 8 & 20 \end{smallmatrix}\right)$ is invertible, so $\operatorname{rank} X = 3$; $(X'X)^{-1} = \frac{1}{4}\left(\begin{smallmatrix} 6 & -1 & -2 \\ -1 & 1 & 0 \\ -2 & 0 & 1 \end{smallmatrix}\right)$, $X'y = (20, 28, 44)'$, $\hat\beta = (1,2,1)'$, $S(\hat\beta) = 4$.

1. Kronecker products and vec

Definition 1. Let $A = (a_{ij}) \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{p\times q}$. The **Kronecker product** of A and B is

$$A \otimes B = \begin{pmatrix} a_{11}B & \cdots & a_{1n}B \\ \vdots & & \vdots \\ a_{m1}B & \cdots & a_{mn}B \end{pmatrix} \in \mathbb{R}^{mp\times nq}$$

with $m \times n$ blocks of size $p \times q$, the (i,j) block $a_{ij}B$: the entry in row $(i-1)p+r$ and column $(j-1)q+s$ is $a_{ij}b_{rs}$. For $a \in \mathbb{R}^m$, $b \in \mathbb{R}^p$: $a \otimes b = (a_1b', \ldots, a_mb')' \in \mathbb{R}^{mp}$. For a number c, $c \otimes B = B \otimes c = cB$. Unlike AB, $A \otimes B$ is defined for any sizes.

Example 2. For $A = (1,\ 2,\ 3)$ and $B = \begin{pmatrix} 6 & 7 \\ 8 & 9 \end{pmatrix}$, compute $A \otimes B$, $B \otimes A$, $I_2 \otimes \mathbf{1}_3$ and $\mathbf{1}_3 \otimes I_2$.

Solution.

Definition 1, block by block: $A \otimes B = [1B \quad 2B \quad 3B]$, $B \otimes A = \left(\begin{smallmatrix} 6A & 7A \\ 8A & 9A \end{smallmatrix}\right)$, $I_2 \otimes \mathbf{1}_3 = \left(\begin{smallmatrix} 1\mathbf{1}_3 & 0\mathbf{1}_3 \\ 0\mathbf{1}_3 & 1\mathbf{1}_3 \end{smallmatrix}\right)$, $\mathbf{1}_3 \otimes I_2 = \left(\begin{smallmatrix} 1I_2 \\ 1I_2 \\ 1I_2 \end{smallmatrix}\right)$:

$$A \otimes B = \begin{pmatrix} 6 & 7 & 12 & 14 & 18 & 21 \\ 8 & 9 & 16 & 18 & 24 & 27 \end{pmatrix}$$

$$B \otimes A = \begin{pmatrix} 6 & 12 & 18 & 7 & 14 & 21 \\ 8 & 16 & 24 & 9 & 18 & 27 \end{pmatrix}$$

$$I_2 \otimes \mathbf{1}_3 = \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{1}_3 \otimes I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Both $A \otimes B$ and $B \otimes A$ contain the twelve numbers a_jb_{rs}, in different orders: $A \otimes B \neq B \otimes A$.

Theorem 3. Let $A = (a_{ij}) \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{p\times q}$.

(a) $(A \otimes B)' = A' \otimes B'$.

(b) (**Mixed product**) For $C = (c_{jk}) \in \mathbb{R}^{n\times s}$ and $D \in \mathbb{R}^{q\times t}$, $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$.

Proof.

(a) $A \otimes B$ has m block rows and n block columns; Lecture 5, Theorem 2(b):

$$\text{block } (j,i) \text{ of } (A \otimes B)' = (a_{ij}B)' = a_{ij}B' = (A')_{ji}B' = \text{block } (j,i) \text{ of } A' \otimes B'$$

(b) $A \otimes B$ has n block columns of width q, $C \otimes D$ has n block rows of height q: conformal for multiplication; Lecture 5, Theorem 2(c), and Lecture 1, Theorem 3(a):

$$\text{block } (i,k) \text{ of } (A \otimes B)(C \otimes D) = \sum_{j=1}^{n}(a_{ij}B)(c_{jk}D) = \Big(\sum_{j=1}^{n} a_{ij}c_{jk}\Big)BD = (AC)_{ik}BD$$

This is block (i,k) of $(AC) \otimes (BD)$, for $i = 1, \ldots, m$, $k = 1, \ldots, s$.

□

Corollary 4. (a) $I_m \otimes I_p = I_{mp}$. For $B \in \mathbb{R}^{p\times q}$ and $D = \operatorname{diag}(d_1, \dots, d_m)$, $D \otimes B$ is block diagonal with the diagonal blocks $d_1B, \dots, d_mB$; in particular $I_m \otimes B$ has m diagonal blocks B.

(b) If $A \in \mathbb{R}^{m\times m}$ and $B \in \mathbb{R}^{p\times p}$ are invertible, then $A \otimes B$ is invertible and $(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}$.

(c) For $A \in \mathbb{R}^{m\times m}$ and $B \in \mathbb{R}^{p\times p}$, $\operatorname{tr}(A \otimes B) = \operatorname{tr} A \cdot \operatorname{tr} B$.

Proof. (a) The (i, j) block of $D \otimes B$ is d_iB for $i = j$ and 0 for $i \neq j$ (Definition 1; Lecture 1, Definition 12); $D = I_m$ gives $I_m \otimes B$, and then $B = I_p$ gives $I_m \otimes I_p = I_{mp}$.

(b) Theorem 3(b); part (a) above:

$$(A \otimes B)(A^{-1} \otimes B^{-1}) = (AA^{-1}) \otimes (BB^{-1}) = I_m \otimes I_p = I_{mp}$$
$$\implies A \otimes B \text{ is invertible, } (A \otimes B)^{-1} = A^{-1} \otimes B^{-1} \qquad \text{(Lecture 2, Theorem 15)}$$

(c) The diagonal blocks of $A \otimes B$ are $a_{11}B, \dots, a_{mm}B$; Lecture 1, Theorem 10(a):

$$\operatorname{tr}(A \otimes B) = \textstyle\sum_{i=1}^m \operatorname{tr}(a_{ii}B) = \left(\sum_{i=1}^m a_{ii}\right) \operatorname{tr} B = \operatorname{tr} A \cdot \operatorname{tr} B$$

□

Definition 5. For $A \in \mathbb{R}^{m\times n}$ with columns $a_1, \dots, a_n$, $\operatorname{vec} A = (a_1', \dots, a_n')' \in \mathbb{R}^{mn}$ stacks the columns: a_{ij} is entry $(j-1)m + i$. For instance $\operatorname{vec} \left(\begin{smallmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{smallmatrix}\right) = (1, 4, 2, 5, 3, 6)'$. For $A, B \in \mathbb{R}^{m\times n}$ and $c, d \in \mathbb{R}$: $\operatorname{vec}(cA + dB) = c \operatorname{vec} A + d \operatorname{vec} B$; $\operatorname{vec} A = \operatorname{vec} B \iff A = B$; $\|\operatorname{vec} A\| = \|A\|_F$ (Lecture 13, Definition 2).

Theorem 6. (a) For $a \in \mathbb{R}^m$ and $b \in \mathbb{R}^n$, $\operatorname{vec}(ab') = b \otimes a$.

(b) For $A \in \mathbb{R}^{m\times n}$, $B \in \mathbb{R}^{n\times p}$ and $C \in \mathbb{R}^{p\times q}$, $\operatorname{vec}(ABC) = (C' \otimes A) \operatorname{vec} B$.

(c) For $A, B \in \mathbb{R}^{m\times n}$, $(\operatorname{vec} A)' \operatorname{vec} B = \operatorname{tr}(A'B)$.

Proof.

(a) Lecture 1, Theorem 4(a); Definition 1:

$$\operatorname{col}_j(ab') = b_j a \implies \operatorname{vec}(ab') = (b_1a', \dots, b_na')' = b \otimes a$$

(b) $b_j = \operatorname{col}_j(B)$, $u_j = \operatorname{col}_j(I_p)$; Lecture 1, Theorems 4(d), 3(a), 8; then (a) and Theorem 3(b); then the sum over j (Definition 5):

$$B = BI_p = \textstyle\sum_{j=1}^p b_j u_j', \qquad ABC = \sum_j (Ab_j)(u_j'C) = \sum_j (Ab_j)(C'u_j)'$$
$$\operatorname{vec}\Big((Ab_j)(C'u_j)'\Big) = (C'u_j) \otimes (Ab_j) = (C' \otimes A)(u_j \otimes b_j) = (C' \otimes A) \operatorname{vec}(b_j u_j')$$
$$\operatorname{vec}(ABC) = (C' \otimes A) \operatorname{vec}\Big(\textstyle\sum_j b_j u_j'\Big) = (C' \otimes A) \operatorname{vec} B$$

(c) Lecture 1, Theorem 4(d); both vectors list the entries in the same order:

$$(A'B)_{jj} = \operatorname{col}_j(A)' \operatorname{col}_j(B) = \textstyle\sum_i a_{ij}b_{ij} \implies \operatorname{tr}(A'B) = \sum_{i,j} a_{ij}b_{ij} = (\operatorname{vec} A)' \operatorname{vec} B$$

□

2. Derivatives and differentials

Definition 7. The **ball** $B(c, \delta)$ ($c \in \mathbb{R}^n$, $\delta > 0$) is the set of $x \in \mathbb{R}^n$ with $\|x - c\| < \delta$; $S \subseteq \mathbb{R}^n$ is **open** if every $c \in S$ has a ball $B(c, \delta) \subseteq S$. For matrices $X, C \in \mathbb{R}^{n\times q}$ the distance is $\|X - C\|_F = \|\operatorname{vec} X - \operatorname{vec} C\|$ (Definition 5): a set of matrices is open if the set of the vectors $\operatorname{vec} X$ is. For a function f on an open S and $x \in S$, the **partial derivative** $\partial f(x)/\partial x_j$ is the derivative at $t = 0$ of

$t \mapsto f(x + te_j)$, when it exists. For $f = (f_1, \dots, f_m)' : S \to \mathbb{R}^m$ with all these partial derivatives at x, the **derivative** (or **Jacobian matrix**) of f at x is

$$\mathrm{D}f(x) = \frac{\partial f(x)}{\partial x'} \in \mathbb{R}^{m \times n} \quad \text{with } (i,j) \text{ entry } \frac{\partial f_i(x)}{\partial x_j}$$

For $m = 1$, $\partial f(x)/\partial x = (\mathrm{D}f(x))'$ is the **gradient**. A matrix function F, with $F(X) \in \mathbb{R}^{m \times p}$, of X in an open set of $\mathbb{R}^{n \times q}$ is the vector function $\operatorname{vec} X \mapsto \operatorname{vec} F(X)$: $\mathrm{D}F(X) = \partial \operatorname{vec} F(X)/\partial (\operatorname{vec} X)' \in \mathbb{R}^{mp \times nq}$. For a scalar function $\varphi(X)$, $\partial\varphi/\partial X \in \mathbb{R}^{n \times q}$ has the (i,j) entry $\partial\varphi/\partial x_{ij}$; so $\mathrm{D}\varphi(X) = \big(\operatorname{vec}(\partial\varphi/\partial X)\big)'$.

Definition 8. Let f be C^1 (F1) on an open $S \subseteq \mathbb{R}^n$, with values in $\mathbb{R}^m$, and $x \in S$. The **differential** of f at x is the linear function $\mathrm{d}f(x) = \mathrm{D}f(x)\,\mathrm{d}x$ of an arbitrary $\mathrm{d}x = (\mathrm{d}x_1, \dots, \mathrm{d}x_n)' \in \mathbb{R}^n$. For a C^1 matrix function F of $X \in \mathbb{R}^{n \times q}$, $\mathrm{d}F(X)$ is the matrix of the size of $F(X)$ with $\operatorname{vec} \mathrm{d}F(X) = \mathrm{D}F(X) \operatorname{vec} \mathrm{d}X$ ($\mathrm{d}X \in \mathbb{R}^{n \times q}$ arbitrary); its (k,l) entry is $\sum_{i,j} (\partial f_{kl}/\partial x_{ij})\,\mathrm{d}x_{ij}$. So $\mathrm{d}f$ has the size of f; for $f(x) = x$, $\mathrm{D}f = I_n$ and $\mathrm{d}f = \mathrm{d}x$.

Theorem 9 (First identification theorem)**.** (a) Let f be C^1 on an open $S \subseteq \mathbb{R}^n$ with values in $\mathbb{R}^m$, $x \in S$ and $A \in \mathbb{R}^{m \times n}$. Then $\mathrm{d}f(x) = A\,\mathrm{d}x$ for every $\mathrm{d}x \iff \mathrm{D}f(x) = A$.

(b) Let F be a C^1 matrix function of $X \in \mathbb{R}^{n \times q}$ with values in $\mathbb{R}^{m \times p}$, and $A \in \mathbb{R}^{mp \times nq}$. Then $\operatorname{vec} \mathrm{d}F(X) = A \operatorname{vec} \mathrm{d}X$ for every $\mathrm{d}X \iff \mathrm{D}F(X) = A$.

(c) Let φ be a C^1 scalar function of $X \in \mathbb{R}^{n \times q}$, and $C \in \mathbb{R}^{n \times q}$. Then $\mathrm{d}\varphi(X) = \operatorname{tr}(C'\mathrm{d}X)$ for every $\mathrm{d}X \iff \partial\varphi/\partial X = C$.

Proof.

(a) Definition 8:

$\Leftarrow$: $\mathrm{d}f(x) = \mathrm{D}f(x)\,\mathrm{d}x = A\,\mathrm{d}x$

$\Rightarrow$: $\mathrm{D}f(x)\,\mathrm{d}x = A\,\mathrm{d}x$ for every $\mathrm{d}x \implies \mathrm{D}f(x) = A$ (Lecture 1, Proposition 11(b))

(b) This is (a) for the vector function $\operatorname{vec} X \mapsto \operatorname{vec} F(X)$ (Definitions 7, 8).

(c) $\operatorname{tr}(C'\mathrm{d}X) = (\operatorname{vec} C)' \operatorname{vec} \mathrm{d}X$ (Theorem 6(c)); (b) with $m = p = 1$, $A = (\operatorname{vec} C)'$; Definition 7:

$$\mathrm{d}\varphi(X) = (\operatorname{vec} C)' \operatorname{vec} \mathrm{d}X \text{ for every } \mathrm{d}X \iff \mathrm{D}\varphi(X) = (\operatorname{vec} C)' \iff \partial\varphi/\partial x_{ij} = c_{ij} \text{ for all } i,j$$

□

So, to find a derivative, compute the differential and bring it to one of these forms.

Theorem 10. Let F, G be C^1 matrix functions (vector and scalar functions included) of the same variable on an open set, A a constant matrix and α a number.

(a) $\mathrm{d}A = 0$, $\mathrm{d}(\alpha F) = \alpha\,\mathrm{d}F$, $\mathrm{d}(F+G) = \mathrm{d}F + \mathrm{d}G$, $\mathrm{d}(F') = (\mathrm{d}F)'$, $\mathrm{d} \operatorname{vec} F = \operatorname{vec} \mathrm{d}F$, and $\mathrm{d} \operatorname{tr} F = \operatorname{tr} \mathrm{d}F$ for square F.

(b) $\mathrm{d}(FG) = (\mathrm{d}F)G + F\,\mathrm{d}G$ whenever FG is defined.

(c) (**Chain rule**) Let $y = g(x)$ be C^1 on an open $S \subseteq \mathbb{R}^n$ with values in an open $T \subseteq \mathbb{R}^m$, and $z = f(y)$ C^1 on T, with $\mathrm{d}z = A(y)\,\mathrm{d}y$ and $\mathrm{d}y = B(x)\,\mathrm{d}x$. Then the differential of $x \mapsto f(g(x))$ is $\mathrm{d}z = A(g(x))B(x)\,\mathrm{d}x$. The same holds for matrix functions, through vec.

Proof. (a) The (k,l) entry of $\mathrm{d}F$ is $\sum_{i,j} (\partial f_{kl}/\partial x_{ij})\,\mathrm{d}x_{ij}$ (Definition 8); the partial derivatives of a constant are 0, those of αf_{kl} and $f_{kl} + g_{kl}$ are $\alpha\,\partial f_{kl}/\partial x_{ij}$ and $\partial f_{kl}/\partial x_{ij} + \partial g_{kl}/\partial x_{ij}$ (F1). F', $\operatorname{vec} F$ and $\operatorname{tr} F$ rearrange or add up entries of F, so their differentials rearrange or add up the differentials of these entries.

(b) The product rule of F1; $(FG)_{kl} = \sum_r f_{kr} g_{rl}$ and (a):

$$\mathrm{d}(uv) = \sum_{i,j} \Big((\partial u/\partial x_{ij}) v + u\, \partial v/\partial x_{ij} \Big) \mathrm{d}x_{ij} = (\mathrm{d}u) v + u\, \mathrm{d}v \qquad (u, v \text{ scalar})$$

$$\mathrm{d}(FG)_{kl} = \sum_r \Big((\mathrm{d}f_{kr}) g_{rl} + f_{kr}\, \mathrm{d}g_{rl} \Big) = \Big((\mathrm{d}F) G + F\, \mathrm{d}G \Big)_{kl}$$

(c) By the chain rule (F1), $\mathrm{D}(f \circ g)(x) = \mathrm{D}f(g(x))\, \mathrm{D}g(x)$, and $\mathrm{D}f(y) = A(y)$, $\mathrm{D}g(x) = B(x)$ (Theorem 9(a)); so the differential of $f \circ g$ at x is $A(g(x))B(x)\, \mathrm{d}x$ (Definition 8). □

Theorem 11. Let $a \in \mathbb{R}^n$ and $A \in \mathbb{R}^{m \times n}$ be constant.

(a) $\mathrm{d}(a'x) = a'\, \mathrm{d}x$: $\mathrm{D}(a'x) = a'$, and the gradient of $a'x$ is a.

(b) $\mathrm{d}(Ax) = A\, \mathrm{d}x$: $\mathrm{D}(Ax) = A$.

(c) For $A \in \mathbb{R}^{n \times n}$, $\mathrm{d}(x'Ax) = x'(A + A')\, \mathrm{d}x$: $\mathrm{D}(x'Ax) = x'(A + A')$, and the gradient is $(A + A')x$, which is $2Ax$ if A is symmetric.

(d) For $X \in \mathbb{R}^{n \times q}$, $a \in \mathbb{R}^n$ and $b \in \mathbb{R}^q$, $\mathrm{d}(a'Xb) = a'(\mathrm{d}X)b$: $\partial(a'Xb)/\partial X = ab'$, and $\mathrm{D}(a'Xb) = (b \otimes a)'$.

Proof.

(a), (b) Theorem 10(a),(b), then Theorem 9(a):

$$\text{(b):}\ \ \mathrm{d}(Ax) = (\mathrm{d}A)x + A\, \mathrm{d}x = A\, \mathrm{d}x$$
$$\text{(a):}\ \ A = a' \text{ in (b)}$$

(c) Theorem 10(a),(b); the number $(\mathrm{d}x)'Ax$ is its own transpose (Lecture 1, Theorem 8):

$$\mathrm{d}(x'Ax) = (\mathrm{d}x)'Ax + x'A\, \mathrm{d}x = x'A'\, \mathrm{d}x + x'A\, \mathrm{d}x = x'(A + A')\, \mathrm{d}x$$

$$\mathrm{D}(x'Ax) = x'(A + A'), \quad \text{gradient } \Big(x'(A + A') \Big)' = (A + A')x \qquad \text{(Theorem 9(a))}$$

(d) Theorem 10(a),(b); Lecture 1, Theorem 10(b):

$$\mathrm{d}(a'Xb) = a'(\mathrm{d}X)b = \mathrm{tr}\Big(a'(\mathrm{d}X)b \Big) = \mathrm{tr}\Big(ba'\, \mathrm{d}X \Big) = \mathrm{tr}\Big((ab')'\, \mathrm{d}X \Big)$$

$$\partial(a'Xb)/\partial X = ab', \qquad \mathrm{D}(a'Xb) = \Big(\mathrm{vec}(ab') \Big)' = (b \otimes a)' \qquad \text{(Theorems 9(c), 6(a))}$$

□

Theorem 12. Let $X \in \mathbb{R}^{n \times n}$ ($n \geqslant 2$) be a variable and $A \in \mathbb{R}^{n \times n}$ constant; c_{ij} are the cofactors of X (Lecture 5, Definition 7).

(a) $\mathrm{d}\,\mathrm{tr}(AX) = \mathrm{tr}(A\, \mathrm{d}X)$, so $\partial\, \mathrm{tr}(AX)/\partial X = A'$.

(b) $\det X$ is a polynomial in the entries of X, $\partial \det X/\partial x_{ij} = c_{ij}$, and $\mathrm{d} \det X = \mathrm{tr}(\mathrm{adj}\, X\, \mathrm{d}X)$. If X is invertible, then $\mathrm{d} \det X = \det X\, \mathrm{tr}(X^{-1}\mathrm{d}X)$ (**Jacobi's formula**) and $\partial \det X/\partial X = \det X\, (X^{-1})'$.

(c) The invertible matrices form an open set, on which $X \mapsto X^{-1}$ is C^2, and $\mathrm{d}X^{-1} = -X^{-1}(\mathrm{d}X)X^{-1}$, $\mathrm{D}(X^{-1}) = -(X^{-1})' \otimes X^{-1}$.

(d) On the open set $\{X : \det X \neq 0\}$, $\mathrm{d} \log|\det X| = \mathrm{tr}(X^{-1}\mathrm{d}X)$; so $\partial \log \det X/\partial X = (X^{-1})'$ where $\det X > 0$.

Proof.

(a) Theorem 10(a),(b); $(A')' = A$:

$$\mathrm{d}\,\mathrm{tr}(AX) = \mathrm{tr}\, \mathrm{d}(AX) = \mathrm{tr}(A\, \mathrm{d}X) = \mathrm{tr}\Big((A')'\mathrm{d}X \Big) \implies \partial\, \mathrm{tr}(AX)/\partial X = A' \qquad \text{(Theorem 9(c))}$$

(b) $\det X$ and its partial derivatives (Lecture 5, Definition 7, Theorem 13(a)):

$\det(x_{11}) = x_{11}$; $\det X = \sum_k x_{1k}(-1)^{1+k} \det X(1|k)$ (row 1)

$\implies \det X$ is a polynomial in the entries, so it is C^2 (induction on the size; F1)

$\det X = \sum_k x_{ik} c_{ik}$, $c_{ik} = (-1)^{i+k} \det X(i|k)$ does not involve row i (row i)

$\implies \det X = c_{ij} x_{ij} + (\text{terms without } x_{ij}) \implies \partial \det X / \partial x_{ij} = c_{ij}$

(b) Definition 8 (1×1 case); $(\operatorname{adj} X)_{ji} = c_{ij}$; if X is invertible, $\det X \neq 0$ and $\operatorname{adj} X = \det X \cdot X^{-1}$ (Lecture 5, Theorems 12(a), 14):

$$\begin{aligned} \mathrm{d} \det X &= \textstyle\sum_{i,j} c_{ij}\, \mathrm{d}x_{ij} = \sum_j \sum_i (\operatorname{adj} X)_{ji}\, \mathrm{d}x_{ij} = \sum_j (\operatorname{adj} X\, \mathrm{d}X)_{jj} = \operatorname{tr}(\operatorname{adj} X\, \mathrm{d}X) \\ &= \det X \operatorname{tr}(X^{-1} \mathrm{d}X) \quad (X \text{ invertible}) \\ \partial \det X / \partial X &= (c_{ij}) = (\operatorname{adj} X)' = \det X\, (X^{-1})' \quad (X \text{ invertible}) \end{aligned}$$

(c) Lecture 5, Theorems 12(a), 14; F1; Theorem 10(a),(b); Theorems 6(b), 9(b):

$\det C \neq 0 \implies \det X \neq 0$ on a ball around C (det continuous, (b))

$X^{-1} = \frac{1}{\det X} \operatorname{adj} X$: polynomials over a nonzero polynomial $\implies C^2$

$X^{-1}X = I_n \implies (\mathrm{d}X^{-1})X + X^{-1}\mathrm{d}X = 0 \implies \mathrm{d}X^{-1} = -X^{-1}(\mathrm{d}X)X^{-1}$

$\operatorname{vec} \mathrm{d}X^{-1} = -\left((X^{-1})' \otimes X^{-1}\right) \operatorname{vec} \mathrm{d}X \implies \mathrm{D}(X^{-1}) = -(X^{-1})' \otimes X^{-1}$

(d) $h(t) = \log t$, $h'(t) = 1/t$ (F1); Theorem 10(c); part (b) above:

$\det X > 0$: $\mathrm{d} \log \det X = \frac{1}{\det X} \mathrm{d} \det X = \operatorname{tr}(X^{-1}\mathrm{d}X)$ ($\{\det X > 0\}$ open: (b), F1)

$\det X < 0$: $\mathrm{d} \log(-\det X) = \frac{1}{-\det X}(-\mathrm{d} \det X) = \operatorname{tr}(X^{-1}\mathrm{d}X)$ ($\{\det X < 0\}$ open)

$\det X > 0$: $\partial \log \det X / \partial X = (X^{-1})'$ (Theorem 9(c), $C = (X^{-1})'$)

□

3. The Hessian matrix and extrema

Definition 13. Let f be a C^2 scalar function on an open $S \subseteq \mathbb{R}^n$. The **Hessian matrix** of f at $x \in S$ is $\mathrm{H}f(x) = \partial^2 f(x) / \partial x\, \partial x' \in \mathbb{R}^{n \times n}$, with (i,j) entry $\partial^2 f(x) / \partial x_i \partial x_j$; it is symmetric (F2), and it is the derivative of the gradient, $\mathrm{H}f = \mathrm{D}\left((\mathrm{D}f)'\right)$. The **second differential** $\mathrm{d}^2 f(x)$ is the differential of $x \mapsto \mathrm{d}f(x) = \mathrm{D}f(x)\, \mathrm{d}x$ with $\mathrm{d}x$ held fixed.

With $a_j = \partial f / \partial x_j$; Theorem 10(a),(b), $\mathrm{d}x$ fixed:

$$\mathrm{d}f = \textstyle\sum_j a_j\, \mathrm{d}x_j, \qquad \mathrm{d}^2 f = \sum_j (\mathrm{d}a_j)\, \mathrm{d}x_j = \sum_j \sum_i \frac{\partial a_j}{\partial x_i}\, \mathrm{d}x_i\, \mathrm{d}x_j = (\mathrm{d}x)'\, \mathrm{H}f(x)\, \mathrm{d}x$$

To compute $\mathrm{d}^2 f$, apply Theorem 10 to $\mathrm{d}f$, with $\mathrm{d}(\mathrm{d}x) = 0$.

Lemma 14. (a) For $A \in \mathbb{R}^{n \times n}$: $x'Ax = 0$ for every $x \in \mathbb{R}^n \iff A' = -A$.

(b) For $A, B \in \mathbb{R}^{n \times n}$: $x'Ax = x'Bx$ for every $x \iff A + A' = B + B'$. For symmetric A, B, $x'Ax = x'Bx$ for every x only if $A = B$.

Proof.

(a) A number is its own transpose; $e_i' A e_j = a_{ij}$ (Lecture 1, Proposition 11(a)):

$\Leftarrow$: $x'Ax = (x'Ax)' = x'A'x = -x'Ax \implies x'Ax = 0$

$\Rightarrow$: $x = e_i$: $a_{ii} = 0$; $x = e_i + e_j$ $(i \neq j)$: $0 = a_{ii} + a_{ij} + a_{ji} + a_{jj} = a_{ij} + a_{ji} \implies A' = -A$

(b) (a) for $A - B$:

$$x'Ax = x'Bx\ \forall x \iff x'(A-B)x = 0\ \forall x \iff (A-B)' = -(A-B) \iff A + A' = B + B'$$

For symmetric A, B this reads $2A = 2B$.

□

Theorem 15 (Second identification theorem)**.** Let f be a C^2 scalar function on an open $S \subseteq \mathbb{R}^n$, $x \in S$ and $B \in \mathbb{R}^{n\times n}$. Then $\mathrm{d}^2 f(x) = (\mathrm{d}x)'B\,\mathrm{d}x$ for every $\mathrm{d}x \iff \mathrm{H}f(x) = \frac{1}{2}(B + B')$.

Proof.

$\mathrm{H}f(x)$ and $\frac{1}{2}(B + B')$ are symmetric; $\mathrm{d}^2 f = (\mathrm{d}x)'\mathrm{H}f\,\mathrm{d}x$ (Definition 13):

$\Rightarrow$: $(\mathrm{d}x)'\mathrm{H}f(x)\,\mathrm{d}x = (\mathrm{d}x)'B\,\mathrm{d}x$ for every $\mathrm{d}x \implies 2\mathrm{H}f(x) = B + B'$ (Lemma 14(b))

$\Leftarrow$: $(\mathrm{d}x)'\frac{1}{2}(B + B')\,\mathrm{d}x = (\mathrm{d}x)'B\,\mathrm{d}x$ ($(\mathrm{d}x)'B'\,\mathrm{d}x = (\mathrm{d}x)'B\,\mathrm{d}x$)

□

For $f(x) = x'Ax$ ($A \in \mathbb{R}^{n\times n}$): $\mathrm{d}f = (\mathrm{d}x)'Ax + x'A\,\mathrm{d}x$, $\mathrm{d}^2 f = 2(\mathrm{d}x)'A\,\mathrm{d}x$, so $\mathrm{H}f = A + A'$, which is $2A$ for symmetric A.

Definition 16. Let f be a scalar function on $S \subseteq \mathbb{R}^n$ and $c \in S$. f has a **local minimum** at c if $f(x) \geqslant f(c)$ for all $x \in S$ in some ball $B(c, \delta)$, a **strict local minimum** if $f(x) > f(c)$ there for $x \neq c$, and a **global minimum** (**strict global minimum**) if $f(x) \geqslant f(c)$ for all $x \in S$ ($f(x) > f(c)$ for $x \neq c$); maxima likewise, with the inequalities reversed. If S is open and f is C^1, c is a **stationary point** of f if $\mathrm{D}f(c) = 0$. S is **convex** if $tx + (1 - t)y \in S$ for all $x, y \in S$ and $0 \leqslant t \leqslant 1$.

Lemma 17. Let $S \subseteq \mathbb{R}^n$ be open, f a C^2 scalar function on S, $c \in S$ and $u \in \mathbb{R}^n$. Let $J = \{t \in \mathbb{R} : c + tu \in S\}$ and $h(t) = f(c + tu)$ for $t \in J$.

(a) J is open and contains 0; h is C^2 on J, with $h'(t) = \mathrm{D}f(c + tu)u$ and $h''(t) = u'\mathrm{H}f(c + tu)u$.

(b) If S is convex and $c+u \in S$, then, for some $0 < \tau < 1$, $f(c+u) = f(c)+\mathrm{D}f(c)u+\frac{1}{2}u'\mathrm{H}f(c+\tau u)u$.

Proof.

(a) $0 \in J$ ($c \in S$); Lecture 6, Definition 1; the chain rule of F1 ($t \mapsto c + tu$ has the derivative u):

$c + t_0 u \in S \implies B(c + t_0 u, \varepsilon) \subseteq S \ni c + tu$ for $|t - t_0|\,\|u\| < \varepsilon \implies J$ open

$h'(t) = \mathrm{D}f(c + tu)u = \sum_j u_j \frac{\partial f}{\partial x_j}(c + tu)$

$h''(t) = \sum_j \sum_i u_j u_i \frac{\partial^2 f}{\partial x_i \partial x_j}(c + tu) = u'\mathrm{H}f(c + tu)u$, continuous in t (Definition 13)

(b) For t between $t_1, t_2 \in J$, $c + tu$ is a convex combination of $c + t_1 u$, $c + t_2 u$:

J is an open interval containing 0 and 1 (S convex; (a))

$f(c + u) = h(1) = h(0) + h'(0) + \frac{1}{2}h''(\tau) = f(c) + \mathrm{D}f(c)u + \frac{1}{2}u'\mathrm{H}f(c + \tau u)u$ (F3; (a))

□

Theorem 18. Let $S \subseteq \mathbb{R}^n$ be open, f a C^2 scalar function on S, and $c \in S$.

(a) (**First-order condition**) If f has a local minimum or maximum at c, then $\mathrm{D}f(c) = 0$.

(b) (**Global minimum**) Let S be convex. If $\mathrm{D}f(c) = 0$ and $\mathrm{H}f(x)$ is PSD for every $x \in S$, then f has a global minimum at c; if $\mathrm{H}f(x)$ is PD for every $x \in S$, it is strict.

(c) (**Constraints**) Let S be convex, g a C^2 function on S with values in $\mathbb{R}^m$, $l \in \mathbb{R}^m$, and $\psi(x) = f(x) - l'g(x)$. If $\mathrm{D}\psi(c) = 0$, $g(c) = 0$ and $\mathrm{H}\psi(x)$ is PSD (PD) for every $x \in S$, then $f(x) \geqslant f(c)$ ($f(x) > f(c)$ for $x \neq c$) for every $x \in S$ with $g(x) = 0$.

Proof.

(a) Local minimum (for a maximum apply this to $-f$): $f(x) \geqslant f(c)$ on a ball $B(c, \delta) \subseteq S$ (δ small; S open); $h(t) = f(c+te_j)$ (Lemma 17, $u = e_j$); $\mathrm{D}f(c)e_j$ is column j of $\mathrm{D}f(c)$ (Lecture 1, Theorem 4(b)):

$c + te_j \in B(c, \delta)$ for $|t| < \delta \implies h$ has a local minimum at $t = 0$

$0 = h'(0) = \mathrm{D}f(c)e_j = \partial f(c)/\partial x_j$ for every $j \implies \mathrm{D}f(c) = 0$ (F3; Lemma 17(a))

(b) $x \in S$, $u = x - c$; Lemma 17(b) (S convex, $c + u = x \in S$) and $\mathrm{D}f(c) = 0$:

$$f(x) = f(c) + \tfrac{1}{2}u'\mathrm{H}f(c + \tau u)u, \qquad c + \tau u = (1 - \tau)c + \tau x \in S$$
$$\mathrm{H}f(c + \tau u) \text{ PSD} \implies f(x) \geqslant f(c); \qquad \text{PD and } x \neq c \ (u \neq 0) \implies f(x) > f(c)$$

(c) ψ is C^2 on S (F1); (b) for ψ:

$$\psi(x) \geqslant \psi(c) \text{ for every } x \in S \ (> \text{ for } x \neq c \text{ in the PD case})$$
$$g(x) = 0 \implies f(x) = \psi(x); \qquad g(c) = 0 \implies \psi(c) = f(c)$$

□

To use (c), solve $\mathrm{D}\psi(x) = 0$, $g(x) = 0$ for x and the **Lagrange multipliers** l (**Lagrange's method**), then check the Hessian of the **Lagrangian** ψ. For linear constraints $g(x) = Rx - r$: $\mathrm{d}^2(l'Rx - l'r) = \mathrm{d}(l'R\,\mathrm{d}x) = 0$, so $\mathrm{H}\psi = \mathrm{H}f$ (Theorem 15).

Example 19. Let $A \in \mathbb{R}^{n\times n}$ be PD, $b \in \mathbb{R}^n$, and $f(x) = \frac{1}{2}x'Ax - b'x$ on $\mathbb{R}^n$. (a) Minimize f. (b) Minimize f subject to $c'x = 0$ ($c \in \mathbb{R}^n$, $c \neq 0$).

Solution. $\mathbb{R}^n$ is open and convex; f is a polynomial, so C^2 (F1).

(a) Theorem 11(a),(c) with $A' = A$; Theorem 15; A is invertible (Lecture 8, Theorem 14(c)), A^{-1} symmetric (Lecture 2, Corollary 12):

$$\mathrm{d}f = x'A\,\mathrm{d}x - b'\,\mathrm{d}x = (Ax - b)'\mathrm{d}x, \quad \mathrm{d}^2 f = (\mathrm{d}x)'A\,\mathrm{d}x, \quad \mathrm{H}f = A \text{ PD at every } x$$
$$\mathrm{D}f(x) = 0 \iff Ax = b \iff x = \hat{x} = A^{-1}b \implies \text{the strict global minimizer} \qquad \text{(Theorem 18(b))}$$
$$(x - \hat{x})'A(x - \hat{x}) = x'Ax - 2b'x + b'A^{-1}b = 2f(x) + b'A^{-1}b \geqslant 0, \ = 0 \text{ at } x = \hat{x}$$
$$\text{minimum: } f(\hat{x}) = -\tfrac{1}{2}b'A^{-1}b$$

(b) $\psi(x) = f(x) - \lambda c'x$; Lecture 5, Theorem 2(c); $A^{-1} = (A^{-1})'AA^{-1}$ is PD (Lecture 8, Theorem 15(a), $K = A^{-1}$, $\operatorname{rank} A^{-1} = n$ by Lecture 5, Theorem 12(a)):

$$\mathrm{d}\psi = (Ax - b - \lambda c)'\mathrm{d}x\colon \ Ax = b + \lambda c, \ c'x = 0 \iff \begin{pmatrix} A & c \\ c' & 0 \end{pmatrix}\begin{pmatrix} x \\ -\lambda \end{pmatrix} = \begin{pmatrix} b \\ 0 \end{pmatrix}$$
$$x = A^{-1}(b + \lambda c), \quad 0 = c'x = c'A^{-1}b + \lambda c'A^{-1}c, \quad c'A^{-1}c > 0$$
$$\tilde{\lambda} = -\frac{c'A^{-1}b}{c'A^{-1}c}, \qquad \tilde{x} = A^{-1}b - \frac{c'A^{-1}b}{c'A^{-1}c}\,A^{-1}c$$
$$\mathrm{H}\psi = \mathrm{H}f = A \text{ PD} \implies \tilde{x} \text{ is the strict minimizer under } c'x = 0 \qquad \text{(Theorem 18(c))}$$

4. Least squares and the Newton–Raphson method

Theorem 20. Let $X \in \mathbb{R}^{n\times p}$, $y \in \mathbb{R}^n$, and $S(\beta) = (y - X\beta)'(y - X\beta)$ for $\beta \in \mathbb{R}^p$.

(a) $\mathrm{d}S = -2(y - X\beta)'X\,\mathrm{d}\beta$, so $\mathrm{D}S(\beta) = -2(y - X\beta)'X$ and the gradient is $2X'X\beta - 2X'y$; $\mathrm{H}S(\beta) = 2X'X$, which is PSD.

(b) $\hat{\beta}$ minimizes $S \iff X'X\hat{\beta} = X'y$ (the normal equations).

(c) If $\operatorname{rank} X = p$, then $\mathrm{H}S$ is PD, and $\hat{\beta} = (X'X)^{-1}X'y$ is the strict global minimizer.

Proof.

(a) $e = y - X\beta$; Theorems 10(a),(b), 11(b); $(\mathrm{d}e)'e = e'\mathrm{d}e$ (a number); $\mathrm{d}\beta$ fixed in d^2S; $X'X$ is PSD (Lecture 8, Theorem 15(b)):

$$\mathrm{d}e = -X\,\mathrm{d}\beta, \quad \mathrm{d}S = (\mathrm{d}e)'e + e'\,\mathrm{d}e = 2e'\,\mathrm{d}e = -2(y - X\beta)'X\,\mathrm{d}\beta$$
$$\mathrm{D}S(\beta) = -2(y - X\beta)'X, \text{ gradient } -2X'(y - X\beta) = 2X'X\beta - 2X'y \qquad \text{(Theorem 9(a))}$$
$$\mathrm{d}^2S = -2(\mathrm{d}e)'X\,\mathrm{d}\beta = 2(\mathrm{d}\beta)'X'X\,\mathrm{d}\beta \implies \mathrm{H}S = 2X'X, \text{ PSD} \qquad \text{(Theorem 15)}$$

(b) S is a polynomial, so C^2 (F1); $\mathbb{R}^p$ is open and convex:

$\Rightarrow$: $\mathrm{D}S(\hat\beta) = 0 \iff X'(y - X\hat\beta) = 0$ (Theorem 18(a))

$\Leftarrow$: $\hat\beta$ is a stationary point, HS PSD everywhere $\implies$ global minimizer (Theorem 18(b))

(c) Lecture 8, Theorem 15(b); Lecture 11, Theorem 2(b):

$\operatorname{rank} X = p \implies X'X$ PD, $x'(2X'X)x > 0$ $(x \neq 0) \implies$ HS PD

$X'X$ invertible, the only solution of the normal equations is $(X'X)^{-1}X'y$

$\implies \hat\beta = (X'X)^{-1}X'y$ is the strict global minimizer (Theorem 18(b))

□

Part (b) is Lecture 11, Theorem 3(a), proved there by projection, without calculus.

Example 21. For X and y of Lecture 11, Example 5, find the gradient of S at $\beta = 0$ and at $\hat\beta = (1, 2, 1)'$, and show that $\hat\beta$ is the strict global minimizer of S.

Solution.

Theorem 20(a),(c): the gradient is $2X'X\beta - 2X'y$; $X'X$, $X'y$, $\operatorname{rank} X = 3$ and $S(\hat\beta) = 4$ from Lecture 11, Example 5:

$$X'X = \begin{pmatrix} 4 & 4 & 8 \\ 4 & 8 & 8 \\ 8 & 8 & 20 \end{pmatrix}, \quad X'y = \begin{pmatrix} 20 \\ 28 \\ 44 \end{pmatrix}: \quad \text{gradient at } \beta = 0 \text{ is } -2X'y = -2(20, 28, 44)' \neq 0$$

$X'X\hat\beta = (4 + 8 + 8,\ 4 + 16 + 8,\ 8 + 16 + 20)' = (20, 28, 44)' = X'y \implies$ gradient at $\hat\beta$ is 0

$\operatorname{rank} X = 3 = p \implies$ H$S = 2X'X$ PD $\implies \hat\beta$ is the strict global minimizer, $S(\hat\beta) = 4$

Example 22 (Newton–Raphson)**.** To solve n equations $f(x) = 0$ in n unknowns (f C^1, $G(x) = \mathrm{D}f(x)$), set the first-order approximation $f(x_r + \Delta) \approx f(x_r) + G(x_r)\Delta$ (the differential at the current x_r) to 0: the **Newton–Raphson** step is

$$x_{r+1} = x_r - G(x_r)^{-1} f(x_r) \qquad (r = 0, 1, 2, \dots)$$

provided $G(x_r)$ is invertible; one stops when the step is negligible. For a stationary point of a C^2 scalar φ: $f = (\mathrm{D}\varphi)'$, $G = \mathrm{H}\varphi$. Show that one step from any x_0 reaches the stationary point for $\varphi(x) = \frac{1}{2}x'Ax - b'x$ (A symmetric and invertible) and for the S of Theorem 20 with $\operatorname{rank} X = p$; take the step from $\beta_0 = 0$ in Example 21.

Solution.

As in Example 19(a), and Theorem 20(a),(c); $(X'X)^{-1}$ from Lecture 11, Example 5:

$(\mathrm{D}\varphi(x))' = Ax - b$, $\mathrm{H}\varphi = A \implies x_1 = x_0 - A^{-1}(Ax_0 - b) = A^{-1}b$, the stationary point

$(\mathrm{D}S(\beta))' = 2X'X\beta - 2X'y$, H$S = 2X'X$ invertible

$\implies \beta_1 = \beta_0 - (2X'X)^{-1}(2X'X\beta_0 - 2X'y) = (X'X)^{-1}X'y$

$$\beta_0 = 0\text{:}\ \ \beta_1 = (X'X)^{-1}X'y = \frac{1}{4}\begin{pmatrix} 6 & -1 & -2 \\ -1 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix}\begin{pmatrix} 20 \\ 28 \\ 44 \end{pmatrix} = \frac{1}{4}\begin{pmatrix} 120 - 28 - 88 \\ -20 + 28 \\ -40 + 44 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} = \hat\beta$$

For other functions several steps are needed.

Practice. The Exercises and Problems of the textbook, Chapter 16: Kronecker products and vec, differentials and derivatives of traces, determinants and inverses, Hessian matrices and extrema, least squares by differentiation, the Newton–Raphson method.